

CO5 – AT1

Course Code: MLA0201

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Course Name: Fundamentals of Machine Learning

Reg.no: 192525002

Constraint-Based Problem

Question 1: Reinforcement Learning for Warehouse Robot Navigation

Problem Understanding

The warehouse robot must reach its destination by moving **Up, Down, Left, or Right**. It must avoid obstacles, use battery efficiently, and reach the destination within **25 moves**. The environment is stochastic, so some actions may not produce the expected movement.

Constraint Analysis

- Only four directions are allowed.
- Obstacles cannot be crossed.
- Each movement consumes one battery unit.
- Maximum battery capacity is **30 units**.
- The destination must be reached within **25 moves**.
- Positive rewards encourage good paths, while blocked paths receive penalties.

Solution Development

I would use **Value Iteration** with an **ϵ -greedy exploration strategy**. Each state represents the robot's position, and actions represent possible movements.

A suitable reward system is:

- Destination reached: **+100**
- Normal movement: **-1**
- Obstacle/blocked movement: **-10**

Value Iteration calculates the expected value of each state and selects the action with the highest long-term reward.

Application and Justification

Value Iteration is suitable because it considers both **rewards and stochastic transitions**. The movement penalty encourages shorter paths, helping the robot conserve battery and reach the destination within the required limit.

Decision: **Value Iteration** provides a practical optimal navigation policy under the given constraints.

Question 2: Multi-Armed Bandit for Online Advertisement Selection Problem

Understanding

The company has **five advertisements** and must select one advertisement for each visitor. Since click probabilities are initially unknown, the system must learn customer preferences while maximizing clicks.

Constraint Analysis

- Five advertisements are available.
- Only one advertisement is shown per visitor.
- Click probabilities are unknown.
- Maximum available interactions: **10,000**.
- Exploration and exploitation must be balanced.
- Cumulative regret should be minimized.

Solution Development

This can be modeled as a **5-Armed Bandit problem**. I would use the **Upper Confidence Bound (UCB)** strategy.

UCB selects advertisements based on:

- Their average reward.
- The uncertainty caused by limited observations.

The reward mechanism is:

- **Click = 1** • **No Click = 0**

After each interaction, the estimated reward for the selected advertisement is updated. **Application and Justification**

Initially, UCB explores advertisements to learn their performance. Later, it increasingly selects advertisements with higher estimated click probabilities.

Decision: UCB is recommended because it balances exploration and exploitation efficiently and helps minimize cumulative regret within the 10,000-interaction budget.

Question 3: Sampling Strategy for Machine Learning Model Development

Problem Understanding

The healthcare company has **one million patient records**, but limited computational resources prevent training on the complete dataset. The model must maintain **above 95% accuracy** and at least **95% confidence**.

Constraint Analysis

- Dataset contains **1,000,000 records**.
- Available memory is limited to **8 GB RAM**.
- All patient categories must be represented.
- Required accuracy: **above 95%**.
- Required confidence: **at least 95%**.
- Sampling bias and computational cost must be minimized.

Solution Development

I would recommend **Stratified Sampling**. The dataset is divided into different patient categories, and proportional samples are selected from every group.

This ensures that important and minority categories are represented in the training data.

Application of Concepts

- **Sample Complexity:** Determines how many records are needed for reliable learning.
- **VC Dimension:** Represents model complexity and influences the amount of data required.
- **Occam's Learning Principle:** Prefers simpler models that generalize well.
- **Accuracy-Confidence Boosting:** More representative data and validation improve reliability.

Justification

Stratified Sampling reduces computational requirements while preserving the original distribution of patient categories. This reduces sampling bias and supports better generalization.

Decision: Stratified Sampling is the most suitable approach because it satisfies the memory limitation while maintaining representative data, high accuracy, and confidence.